

Security-focused researcher and former cybersecurity consultant specializing in AI agent security and LLM integrations. Aiming to apply deep technical knowledge of the Model Context Protocol (MCP), LangGraph, and zero-trust architectures to build, orchestrate, and govern secure AI systems.

EDUCATION

Ph.D. Electrical and Computer Engineering, Georgia Institute of Technology (GPA: 3.83/4)	July 2028 (Expected)
M.S. Electrical and Computer Engineering, Georgia Institute of Technology (GPA: 3.83/4)	Dec 2026
M.Tech. (Res.) Electrical Communication Engineering, Indian Institute of Science, Bangalore (GPA: 8.3/10)	July 2023
M.S. (Hons.) Economics, BITS Pilani - Goa Campus (GPA: 8.01/10)	July 2017
B.E. (Hons.) Electrical and Electronics Engineering, BITS Pilani - Goa Campus (GPA: 8.01/10)	July 2017

COURSES

Georgia Tech	Conversational AI, Statistical Machine Learning, Probabilistic Graphical Models in Machine Learning, Convex Optimization, Machine Learning Security, Computer Network Security, Advanced Malware Analysis, Introduction to Information Security, Computer Communication Networks
Udemy	LangChain, LangGraph, Model Context Protocol, AI Agent Security

SKILLS

AI Security Tools	PyRIT, Garak, Promptfoo, TextAttack, Burp Suite, Sengrep, Wireshark
Languages & Frameworks	Python, Bash, LangChain, LangGraph, Model Context Protocol (MCP), AutoGen, Transformers
Data & MLOps Infra	Pinecone, ChromaDB, Milvus, MLflow, Kubernetes, Docker, Triton Inference Server, AWS/Azure AI
AI/LLM Offensive Tactics	Prompt Injection (Direct/Indirect), Jailbreaking, Agent Goal Hijacking, RAG Exfiltration, Prompt Leaking
Model & Data Attacks	Data Poisoning, Model Inversion, Insecure Output Handling, Supply-Chain Tampering, Evasion
Compliance & Governance	OWASP Top 10 for LLMs, NIST AI RMF, MITRE ATLAS, ISO/IEC 42001
Vulnerability Scanning	Nessus

TECHNICAL EXPERIENCE

Graduate Research Assistant Georgia Institute of Technology	Aug 2023 — Present
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Cybersecurity Consultant EY, Bangalore	Jul 2017 — Aug 2020
Offensive Security Testing – Corporate Network	
- Breached into and compromised corporate domain of at least 5 multinational companies. - Gained administrative access to the domain controller, the ‘crown jewel’ of the corporate network. - Created backdoor domain administrators for few, with potential to spread ransomware to entire domain. - For 2 of these companies, the entire compromise was performed completely remote with zero initial knowledge.	

RELEVANT PUBLICATIONS

BackDelta: Backdoor Direction Erasure via Low-rank Task Arithmetic in Large Language Models	2026
Submitted to Network and Distributed System Security Symposium (NDSS), 2027	
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PROJECTS

Secure Autonomous HR Agent   	2026
- Architected an autonomous HR agent leveraging LangChain and Model Context Protocol (MCP) to autonomously screen candidate resumes against specific job descriptions. - Constructed comprehensive proof-of-concept exploits covering the OWASP Top 10 for LLMs, demonstrating critical vectors including indirect prompt injection, insecure output handling, and excessive agency. - Performed comprehensive threat modeling using MITRE ATLAS on the AI agent architecture to uncover vulnerabilities related to prompt injection and RAG exfiltration. - Secured the application pipeline using semantic guardrails and input-filtering frameworks to intercept jailbreaking attempts and goal hijacking vectors.	

(346) 631-6156
Atlanta, GA
varkeymjohn@gmail.com

Varkey M. John

varkeymjohn.github.io
github.com/varkeymjohn
linkedin.com/in/varkeymjohn

Zero-Trust MCP Security Gateway & Observability Proxy    2026

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LangGraph-Orchestrated AI Vulnerability Scanner for CI/CD    2026

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CERTIFICATIONS

AWS Certified Security Specialty (AWS)	2026
Certified AI Security Professional (Practical DevSecOps)	2026
Applied Time Series Forecasting (Data Science with Marco)	2026
Post Graduate Programme in Internet of Things (BITS Pilani)	2018

ACTIVITIES

Chairperson, IEEE-IISc Communications Society, IISc, Bangalore	2021 — 2022
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AWARDS AND ACHIEVEMENTS

Student fellowship grant, Cisco Centre for Networked Intelligence, IISc Bangalore
Membership in IEEE Eta Kappa Nu
‘Best Officer Award’ and ‘Outstanding Volunteer Award’, IEEE-IISc, Bangalore
‘Spotlight Award’ for ‘Quality Expert’ (awarded twice), EY, Bangalore